

Final Report — The Impact of Daily Notifications on Mental Recovery Across Age Groups

Course: DSA 210 — Introduction to Data Science (Spring 2026) **Author:** Ufuk Çimen **Repository:** <https://github.com/ufukcimen/dsa210-project>

1. Motivation

In the modern digital environment, smartphone notifications constantly interrupt attention and may impose cognitive load that interferes with mental recovery (sleep quality, fatigue, stress regulation). Two questions motivate this study:

1. Does higher notification exposure reduce mental recovery?
2. Are younger individuals more sensitive to digital interruptions than older ones?

If notifications harm recovery, the implication is practical and personal: capping or batching notifications could become a low-cost lifestyle intervention. If age modifies the effect, the policy and product-design implications shift toward age-targeted advice (e.g., focus-mode defaults for younger users).

2. Data Source

The project uses the publicly available Kaggle dataset **Sleep, Screen Time, and Stress Analysis**: <https://www.kaggle.com/code/lukhilaksh/sleep-screen-time-and-stress-analysis/input>

Property	Value
Rows	15,000 individuals
Columns	13
Format	CSV
Missing values	0
Duplicates	0

Variables used:

Variable	Type	Range
age	int	18–59
gender	str	Female / Male / Other
occupation	str	8 categories
daily_screen_time_hours	float	1.0–10.0
phone_usage_before_sleep_minutes	int	0–119
sleep_duration_hours	float	4.0–9.0
sleep_quality_score	float	1.0–10.0
stress_level	float	1.0–10.0
caffeine_intake_cups	int	0–4
physical_activity_minutes	int	0–119
notifications_received_per_day	int	20–299
mental_fatigue_score	float	1.0–10.0

The dataset was downloaded directly from Kaggle, imported with `pandas`, and committed to the repository under `data/raw/`.

3. Data Analysis

The full pipeline is implemented across two notebooks:

- `notebooks/eda_hypothesis_tests.ipynb` — cleaning, feature engineering, EDA, hypothesis testing.
- `notebooks/ml_models.ipynb` — supervised machine learning (regression + classification).

3.1 Data Cleaning

- Column names standardized (lowercase, underscores).
- No duplicates removed (none present).
- No missing values imputed (none present).
- Final shape after cleaning: **15,000 × 13**.

3.2 Feature Engineering

Baseline Mental Recovery Index (MRI)

Min-max normalize each component to [0, 1], then:

```
MRI_additive = (sleep_quality_norm + sleep_duration_norm  
                - mental_fatigue_norm - stress_level_norm) / 4
```

Higher MRI $\Rightarrow$ better mental recovery.

Alternative MRI Variants

To check that conclusions are robust to the choice of formula and to the way normalization handles outliers, three additional variants were constructed:

Variant	Formula	Why
mri_multiplicative	$(\text{sleep_quality} \times \text{sleep_duration}) / (\text{fatigue} \times \text{stress})$	Multiplicative interactions; preserves outliers (no compression).
mri_zscore	additive formula on z-score-normalized components	Unbounded, preserves outlier magnitude.
mri_weighted	additive with weights 0.35 / 0.25 / 0.20 / 0.20	Emphasizes sleep quality as the dominant recovery driver.

Result: The four variants are highly correlated (Spearman $r > 0.9$ between most pairs). Pearson correlation with notifications stays near zero **regardless** of which MRI variant is used — confirming the null finding for notifications is not an artifact of formulation.

Outlier Sensitivity Check

- IQR-based outlier detection (1.5-IQR rule) on each component and on each MRI variant.
- Hypothesis tests re-run on the **outlier-free subset** for each MRI variant.
- **All correlations stayed near zero** after removing outliers — normalization is not hiding meaningful outlier-driven signal.

Grouping Variables

- `age_group` — bins [10, 20, 35, 50, 70] with labels ['10-20', '20-35', '35-50', '50-70'].
- `notification_group` — `pd.qcut` into 3 quantile-based groups: Low / Medium / High.

Processed Data

The cleaned and enriched dataset is saved as `data/processed/processed_dataset.csv` (15,000 × 20).

3.3 Exploratory Data Analysis

The EDA notebook produces:

- Distributions (histograms) of notifications and MRI.
- Scatter plot of notifications vs MRI with regression line.
- Boxplots of MRI and notifications by age group.
- Correlation heatmap of all numeric variables.
- Grouped boxplot of MRI by age × notification group.
- Comparison plots for the four MRI variants and an outlier-sensitivity table.

3.4 Hypothesis Testing

Three classical tests were performed at $\alpha = 0.05$:

Test	H0	Statistic	p-value	Decision
1. Pearson correlation (notifications vs MRI)	$r = 0$	$r = 0.006$	0.495	Fail to reject H0
2. One-way ANOVA (MRI across age groups)	$\mu_1 = \mu_2 = \mu_3 = \mu_4$	$F = 0.34$	0.799	Fail to reject H0
3. Independent t-test (Low vs High notifications)	$\mu_{\text{Low}} = \mu_{\text{High}}$	$t = -0.62$	0.535	Fail to reject H0

None of the three tests detected a significant effect involving notifications or age.

3.5 Machine Learning

Two supervised tasks were tackled to investigate predictability beyond simple correlations.

Setup

- 80 / 20 train-test split, `random_state = 42`.
- 5-fold cross-validation for robust estimates.
- One-hot encoding for `gender` and `occupation`.
- `StandardScaler` for linear models; raw features for tree models.
- **Target leakage prevented** by excluding the four MRI components and all MRI variants from the feature set.

Regression: predict continuous MRI

Model	Test R ²	CV R ²	RMSE	MAE
Linear Regression	0.690	0.680	0.114	0.092
Ridge Regression	0.690	0.680	0.114	0.092
Random Forest	0.690	0.684	0.114	0.092
Gradient Boosting	0.699	0.692	0.113	0.090

All models explain roughly **70% of the variance in MRI** using only the non-MRI behavioral and demographic features.

Classification: high vs low MRI (median split)

Model	Accuracy	CV Accuracy	F1	ROC-AUC
Logistic Regression	0.850	0.855	0.851	0.937
Random Forest	0.848	0.854	0.849	0.930
Gradient Boosting	0.851	0.853	0.853	0.935

Models achieve $\approx$ **85% accuracy** and **ROC-AUC $\approx$ 0.93**, far above the 0.50 random baseline.

Feature Importance

Random Forest gain importance:

Feature	Importance
daily_screen_time_hours	0.84
phone_usage_before_sleep_minutes	0.07
notifications_received_per_day	0.03
physical_activity_minutes	0.03
age	0.02
all others combined	< 0.02

Random Forest permutation importance (drop in accuracy when feature is shuffled):

Feature	Δ accuracy
daily_screen_time_hours	-0.341
phone_usage_before_sleep_minutes	-0.012
notifications_received_per_day	≈ 0
age	≈ 0

`daily_screen_time_hours` carries essentially all the predictive signal.

4. Key Findings

1. **Daily screen time is the dominant predictor of mental recovery.** It has a Pearson correlation of $r \approx -0.80$ with MRI and accounts for $\approx 84\%$ of the Random Forest's feature importance. Models built from this variable alone explain most of the regression performance.
2. **Pre-sleep phone usage is a secondary but real predictor** ($r \approx -0.19$; $\approx 7\%$ feature importance). Heavy phone use right before bed is associated with lower recovery, independently of total screen time.
3. **The original hypothesis about notifications is not supported.** The raw notification count has **no statistically significant relationship with MRI** ($r \approx 0.006$, $p \approx 0.50$), no significant t-test difference between Low and High notification groups, and negligible feature importance ($\approx 2.6\%$ in trees, near-zero permutation importance).
4. **Age does not modify mental recovery.** One-way ANOVA across age groups is non-significant ($F = 0.34$, $p = 0.80$); age has $\approx 2\%$ feature importance in ML models.
5. **Robustness check:** All four alternative MRI computations (additive min-max, multiplicative, z-score, weighted) and the outlier-sensitivity analysis agree — the null finding for notifications is real, not an artifact of normalization compressing outliers.
6. **Machine learning** quantifies the same story: high R^2 and accuracy are achievable, but they come entirely from screen time and pre-sleep phone use, not from notification volume or age.

Interpretation

The project's headline takeaway shifts: **the volume of notifications is the wrong proxy for digital interruption**. What matters for mental recovery is **how much total time** people spend on their devices and **how late** they use them — not the raw number of pings they receive. This nuance is important for product designers and personal-habit guidance: capping notifications without reducing overall screen time may not improve sleep or fatigue.

5. Limitations and Future Work

Limitations

- **Cross-sectional data.** Even with strong correlations, we cannot infer causation. Screen time may *cause* poor sleep, or both may share an unmeasured common cause (e.g., chronic stress, occupation, lifestyle).
- **Likely simulated dataset.** Perfectly balanced categorical distributions and very uniform numeric ranges suggest the data is synthetic. The screen-time signal is realistic and consistent with the broader sleep-and-screens literature, but exact effect sizes should not be over-interpreted as real-world coefficients.
- **MRI is a custom composite**, not a clinically validated instrument. Conclusions are about this constructed index, not a recognized psychological measure.
- **Coarse measurement of notifications.** The dataset records only the *count* of notifications, with no information on type (work / social / spam), urgency, time of day, or whether the user actually attended to them. Real-world impact likely depends on these characteristics, which are invisible here.

- **Self-reported variables** (sleep, stress, fatigue) are subject to recall and social-desirability bias.

Future Work

1. **Richer notification data.** Repeat the analysis with notification *type*, *time-of-day*, *urgency*, and *response behavior* — the count alone misses the construct.
2. **Mediation analysis.** Test whether the effect of screen time on MRI is mediated by phone-usage-before-sleep, since these two variables likely sit on a causal chain.
3. **Non-linear and interaction modeling.** Explore whether the screen-time effect varies by occupation or gender using interaction terms in regression or SHAP analysis of tree models.
4. **Longitudinal / experimental design.** A within-person diary study (or randomized notification-reduction trial) would be necessary to establish causation.
5. **External replication.** Apply the same pipeline to a different, ideally real-world dataset (e.g., RescueTime data, app-usage logs paired with wearable sleep data) to test whether the screen-time-dominates-notifications conclusion generalizes.

6. Reproducibility

- All code is in Python 3, organized in two notebooks under `notebooks/`.
- Dependencies are pinned in `requirements.txt`.
- `random_state = 42` is set for all ML splits and models.
- To reproduce:

```
pip install -r requirements.txt
jupyter notebook notebooks/eda_hypothesis_tests.ipynb      # run all cells
jupyter notebook notebooks/ml_models.ipynb                 # run all cells
```

- The processed dataset (`data/processed/processed_dataset.csv`) is regenerated each time the EDA notebook is run.

7. AI Tool Disclosure

In some parts of this project, AI was used as a helper.

- **Tool:** Anthropic Claude (accessed via Claude Code CLI).
- **Areas where AI was used, with sample prompts:**
 - **Looking up library syntax and small code snippets.**
 - "What's the correct syntax for `pd.qcut` to make three quantile-based groups labeled Low / Medium / High?"
 - "How do I get feature importances out of a `RandomForestRegressor` in `scikit-learn`?"

- **Sanity-checking alternative MRI formulations before implementing them.**
 - *"Does $(\text{sleep_quality} \times \text{sleep_duration}) / (\text{mental_fatigue} \times \text{stress_level})$ make sense as a multiplicative MRI formula?"*
 - *"What's the difference between min-max and z-score normalization in terms of how they treat outliers?"*
 - **Conceptual questions about statistics and ML.**
 - *"Why might a Random Forest still find predictive signal in features whose Pearson correlation with the target looks near zero?"*
 - *"Is one-way ANOVA the right test for comparing MRI means across four age groups?"*
 - **Asking for ways to extend the ML notebook.**
 - *"How can I improve my ML notebook? It has Linear Regression, Logistic Regression, Random Forest, and Gradient Boosting."*
 - Claude recommended adding Ridge regression, MAE, ROC-AUC, and permutation importance, which were then added.
 - **Light copy-editing on a few markdown explanations and parts of this report.**
 - *"Can you make this paragraph shorter and clearer?"*
-

8. Repository Layout

```
dsa210-project/  
├─ README.md  
├─ FINAL_REPORT.pdf  
├─ proposal.pdf  
├─ requirements.txt  
├─ .gitignore  
├─ data/  
│   ├─ raw/  
│   │   └─ sleep_mobile_stress_dataset_15000.csv  
│   └─ processed/  
│       └─ processed_dataset.csv  
└─ notebooks/  
    ├─ eda_hypothesis_tests.ipynb  
    └─ ml_models.ipynb
```